The Future of EEG Sonification: Applications in Rehabilitation, Accessibility, and Human–Computer Interaction

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Abstract

Electroencephalography (EEG) has traditionally been used to diagnose and monitor neurological conditions such as epilepsy. EEG sonification, the translation of neural activity into non-speech sound, offers a way to extend its use beyond the clinic. This paper asks to what extent EEG sonification can expand EEG into rehabilitation, accessibility, and human–computer interaction. A narrative review of literature identified through PubMed examines four application areas: music-based motor rehabilitation, EEG neurofeedback, brain–computer music interfaces (BCMIs), and wearable and monitoring technologies. The review is supported by an exploratory Python prototype that synthesised sine, square, triangle, and sawtooth waveforms, shaped them with an attack–decay–sustain–release envelope, and used an 8 Hz oscillation to modulate the amplitude of an eight-note pentatonic melody. The literature suggests that the evidence varies by application. BCMIs make the strongest case for accessibility, giving people with severe motor disabilities a means of creative expression. Music-based rehabilitation is the most clinically developed, but its effects vary across studies. Neurofeedback and consumer wearables show mixed and less reliable results. The prototype shows that the same signal can produce very different sounds depending on which musical parameter it controls, which highlights the lack of

a standardised mapping method. Across all applications, the main limitations are variability between individuals in EEG signals and concerns about mental privacy. EEG sonification is therefore best understood, at present, as a complement to diagnostic EEG rather than a replacement for it.

Keywords: EEG; sonification; brain–computer music interface; neurofeedback; music therapy; Python

Introduction

Music has long been seen as something created to influence humans, whether by changing people's mood, evoking a certain feeling, or shaping how humans perceive and respond to the world around them (Schafer et al., 2013). From calming melodies that reduce stress, such as lullabies that put children to sleep, to energetic rhythms that encourage movement, music can produce remarkably diverse effects on the human brain (Zaatar et al., 2023).

Electroencephalography (EEG) sonification inverts that relationship. Rather than observing how the brain responds to music, the brain itself becomes the source of the music. This is not meant literally, as brainwaves are not the direct equivalent of sound waves; rather, the oscillating electrical activity in the cerebral cortex is taken and translated into auditory terms such as pitch and tempo.

The pairing of EEG with sound is not arbitrary (Hermann et al., 2011). Sound is well suited to representing many kinds of data, as listeners can track changes in pitch, rhythm, and timbre intuitively without specialist training (Bornmann 2024). Because the auditory system is highly sensitive to changes over time, some patterns may be easier to detect by ear than visually on a screen (King, 2009). EEG–music systems have therefore been proposed as a complementary way

of communicating brain states, whether to support a clinician assessing a patient's condition or to give a patient a means of self-expression (Gupta, 2026).

This paper addresses the following research question: To what extent can EEG sonification expand the use of electroencephalography beyond traditional neurological diagnosis into rehabilitation, accessibility, and interactive music systems?

EEG technology has traditionally been used as a diagnostic and monitoring tool, especially for neurological conditions such as epilepsy and sleep disorders (Rayi et al., 2022). However, as advances in signal processing and hardware enable wearable technology and brain-computer interfaces, EEG use is gradually developing beyond diagnosis alone (Liu et al., 2025).

Examining these possibilities is important because EEG sonification could make neural activity more accessible and interactive, giving patients potential new forms of rehabilitation and creative expression. Its limitations also warrant examination, given the variability of EEG signals across individuals and recording sessions, which may limit how reliable and widely adopted these technologies can become.

This paper presents a narrative review of the practical and clinical applications of EEG sonification, supported by an illustrative Python sonification prototype. In particular, it examines how translating neural activity into sound could contribute to four areas: music-based motor rehabilitation, EEG neurofeedback, brain–computer music interfaces for individuals with disabilities, and wearable technologies capable of producing sound responsive to changes in brain activity and physiological states. The remainder of the paper is organised as follows. The Literature Review summarises the neurophysiological background and the evidence for each application; the Methodology describes how the prototype was developed; the Results and

Discussion evaluate the prototype output and the evidence across applications; and the Conclusion answers the research question.

Literature Review

EEG electrodes are placed on the scalp to record voltage fluctuations produced by ionic currents in neurons. These fluctuations arise primarily from the summed postsynaptic activity of large populations of synchronously active pyramidal neurons in the cortex (Buzsaki et al., 2012). Unlike functional magnetic resonance imaging (fMRI), which measures a slow haemodynamic response, EEG reflects neuronal electrical activity directly and with millisecond temporal resolution (Nunez et al., 2006). Together with its relatively low cost and portability, this makes EEG particularly suitable for real-time applications such as music generation and neurofeedback.

The raw EEG signal consists of different frequency bands, each of which has been associated with particular cognitive states. Delta (0.5–4 Hz) is prominent in deep sleep, while theta (4–8 Hz) is associated with drowsiness and, in some studies, creativity. Alpha (8–13 Hz) is associated with relaxed wakefulness; beta (13–30 Hz) with active thinking and focus; and gamma (above 30 Hz) with higher-order cognitive processing (Newson et al., 2019).

These frequency bands provide a useful framework for interpreting EEG signals, but they should not be treated as definitive indicators of a particular mental state. EEG activity can vary depending on the recording site, individual, task, and other conditions (Newson et al., 2019). Therefore, frequency bands do not map one-to-one onto particular mental states; they are broad indicators that need more specific, situational evidence to be accurately interpreted.

Sonification is the use of non-speech audio to convey information or represent data (Kramer et al., 1999). Beyond the intuitive tracking of pitch, rhythm, and timbre noted above, listening allows a clinician to monitor a signal without watching a screen and allows a patient to hear their own brain activity. This dual role is what links EEG as a clinical tool to EEG as a therapeutic one. Because no single standardized method exists for mapping neural activity onto musical parameters, the mapping chosen depends on the purpose of the system (Ziemann et al., 2026).

Currently, the most clinically developed related application is music-based therapy for patient recovery after stroke and other brain injuries (Xu et al., 2022). In this form of therapy, patients practice finger or arm movements on an instrument or sonification device so that each movement produces a musical note. Strictly speaking, this is movement sonification rather than EEG sonification: the sound is driven by the patient's movements, and EEG is used to measure the resulting neural changes. Studies of this therapeutic approach have found increases in EEG connectivity between sensorimotor and auditory cortices alongside improvements in motor timing (Altenmüller et al., 2009), although the magnitude and consistency of these effects vary across studies (Scholz et al., 2016).

A second application is EEG neurofeedback, in which a person's brain activity is turned into sounds or music that provide immediate feedback (Marzbani et al., 2016). When the brain shows the target pattern, the person receives a response instantly, encouraging them to repeat that brain state. Research on this approach has shown mixed results. Some individual studies have reported improvements in attention, particularly in children with attention-deficit/hyperactivity disorder (ADHD) (Shereena et al., 2019). However, meta-analyses have found that these improvements are less consistent when assessed on a broader scale. Cortese et al. (2016), for example, found no significant benefits to ADHD symptoms when analyzing blinded assessments. Similarly, another

recent meta-analysis of 38 different trials found no meaningful overall improvement in ADHD symptoms, although small-scale individual benefits must be noted (Westwood et al., 2025). There is some evidence that music-based feedback is more engaging than simple tones (Sayal et al., 2025), and one study of school children reported improvements in creative music performance following neurofeedback training (Gruzelier et al., 2014). Overall, this type of intervention appears to have much potential for further investigation, although its benefits are still being studied and results may vary between individuals.

A third application is the brain–computer music interface (BCMI), which allows people who cannot move or speak to compose and perform music using brain activity alone. This may include people with paralysis or other severe motor disabilities, including locked-in syndrome (Miranda et al., 2011). An experiment was conducted where a system presented patients with severe motor impairments with flashing options representing musical phrases (Eaton et al., 2015). The system detected which option a patient was attending to from the visual cortex response recorded by EEG electrodes (steady-state visual evoked potentials), and the selected phrase was then performed by live musicians. Participating patients reportedly described the experience as highly rewarding, giving them back a sense of control that they had lost with their disability and giving them a way to interact with others through music. Although this work does not aim to improve patients' physical condition, it addresses something rarely considered in assistive technology: restoring a degree of creative agency to people whose opportunities for self-expression are severely limited. Because such systems usually rely on visual attention, however, further research is needed to give people without reliable eye control access to musical expression.

A fourth area of application is wearable and monitoring technology. One line of research converts EEG patterns into sound so that patterns can be recognized by ear rather than by eye, helping clinicians and untrained listeners identify abnormal activity without reading a visual trace. When EEG recordings containing seizures were sonified, listeners with little or no EEG training rapidly learned to distinguish seizure from non-seizure activity (Loui et al., 2014; Parvizi et al., 2018). This suggests potential for supporting seizure monitoring at home or at the bedside by non-specialists, although such tools would complement rather than replace clinical diagnosis. In addition, low-cost consumer EEG headsets such as Muse and Emotiv have been used to provide EEG-driven audio feedback for meditation and stress management, estimating the wearer's approximate mental state and adjusting the sound accordingly. For example, Bhayee et al. (2016) used a Muse EEG headband to deliver real-time auditory feedback during a mindfulness training program, with participants receiving changing sounds based on their EEG activity. Several studies report encouraging results, but the precision and reliability of these devices remain limited, with EEG signals varying from person to person and therefore making it difficult to pinpoint the exact mental state of the user (Mikhaylov et al., 2024).

Across these four applications, three recurring issues emerge. First, the evidence is inconsistent: neurofeedback outcomes in particular range from clear improvement to temporary or insignificant effects. Second, individual variability limits generalization, because one person's results may not apply to the next. Third, EEG data can reveal private cognitive information, and many users may be reluctant to share it with consumer devices, especially when early data are collected for research (Ienca & Andorno 2017). In addition, BCMI research still depends largely on visual control, leaving people with more severe disabilities without access.

Methodology

Sonification prototype

To examine how mapping decisions shape EEG sonification in practice, the author developed an exploratory sonification prototype in Python. The prototype was designed around the four stages commonly used in EEG sonification pipelines: preprocessing, feature extraction, parameter mapping, and sound synthesis. Because the aim at this stage was to investigate the synthesis and mapping stages rather than to analyze real neural data, sample recorded EEG data was used, and the preprocessing stage (filtering and artefact removal) was implemented. EEG oscillations of interest (approximately 0.5–100 Hz) lie mostly below or at the lower edge of the human audible range (approximately 20 Hz–20 kHz), so the author adopted a parameter-mapping approach, in which a low-frequency signal controls properties of a separate audible tone, rather than audification, in which the EEG itself is time-compressed or frequency-shifted into the audible range (Vialatte et al., 2012).

The prototype was written using NumPy for numerical processing, SciPy for signal processing, and Matplotlib for visualisation, and audio was written to WAV files using scipy.io.wavfile / played back using sounddevice at a sampling rate of 44.1 kHz. The author first generated four basic waveforms at 440 Hz: sine, square, triangle, and sawtooth. These were chosen because each has a different harmonic content and therefore a distinct timbre. The sine wave is a pure tone with no harmonics; the square wave contains only odd harmonics, giving a hollow, buzzy quality; the triangle wave contains the same odd harmonics, but they decay much faster, giving a softer tone closer to the sine wave; and the sawtooth wave contains all harmonics, making it the brightest of the four. Each waveform was plotted to check its shape.

Next, the author applied an attack–decay–sustain–release (ADSR) envelope to each waveform to shape how its amplitude changed over time. The envelope rose from silence to its peak over an attack of 0.1 s, fell to a sustain level of 0.7 over a decay of 0.2 s, held that level for the remainder of the note, and faded to silence over a release of 0.3 s. This step turned a flat, continuous tone into a sound with a clear onset and ending, closer to a musical note.

The author then combined these components into an eight-note melody to test a simple mapping. An 8 Hz sine-wave oscillation, chosen to stand in for brain activity at the boundary of the theta and alpha bands, modulated the amplitude of each note. Pitches were drawn at random from a C major pentatonic scale, and timbre was set by the choice of waveform. In parallel, the author wrote a feature-extraction function that computes band power (using Welch's method), spectral entropy, root-mean-square (RMS) amplitude, and zero-crossing rate from a signal segment, as candidates for driving musical parameters in later versions; this function was not applied to recorded EEG or connected to the sound. To illustrate how such features could be compared across mental states, the author also produced a violin plot and a correlation heatmap from hand-specified example band-power values for "calm", "focused", and "stressed" conditions; these values were not derived from recorded EEG. The mapping was kept deliberately simple, because its purpose was to show how design choices shape the resulting sound, not to provide a validated sonification method.

Results and Discussion

The synthesis programs produced the expected waveform behavior. The sine, square, triangle, and sawtooth waves at 440 Hz had visibly different shapes (Figure 1), matching the timbral differences described in the methodology. Applying the ADSR envelope turned a steady tone into a note-like sound, with a gradual rise over 0.1 s, a drop to the sustain level, and a fade-out

(Figure 2). In the eight-note melody, the 8 Hz modulation made each note pulse eight times per second.

Two limitations of these outputs shape how they should be read. First, the program did not use recorded EEG: the oscillation was a synthetic sine wave, and the band-power values used in the violin plot (Figure 3) and correlation heatmap (Figure 4) were written by hand for illustration. The class differences in those plots were built into the numbers, so they show how such features could be visualized, not that real EEG differs between "calm," "focused," and "stressed" states. Second, the melody carried no information about brain activity, because pitch was chosen at random and only the modulation rate stood in for a brainwave. The feature extraction function (band power, spectral entropy, RMS, zero crossings) was not connected to the sound.

Even so, the program shows something relevant to the research question. Many different sounds could be made from the same underlying signal depending on which parameter it controls (pitch, modulation rate, or timbre), and nothing in the signal dictates that choice. This is the lack of a standardized mapping described in the literature review, shown in practice: a listener cannot tell whether a given sound reflects the brain or the designer's decisions unless the mapping is stated.

The literature suggests that the evidence differs by application. Brain-computer music interfaces make the clearest case for expanding EEG beyond diagnosis, because they offer people who cannot move or speak a form of creative expression that few assistive technologies address. Music-based motor rehabilitation has the most clinical development, but its effects vary across studies. Neurofeedback has shown mixed results, although music appears to make the process more engaging than simple tones (Sayal et al., 2025). Wearables are the most accessible but the least reliable, since low-cost devices struggle to pinpoint a user's mental state. Seen alongside the

program, one possible interpretation of this pattern is that the applications that work best are those where the mapping is simple and the user knows what the sound means (choosing a flashing icon, moving an arm to play a note), and the ones that struggle depend on inferring a mental state from the signal.

While EEG sonification shows potential for future applications and improvements, several significant drawbacks apply to each application. EEG signals can vary significantly across individuals, so one person's results in an experiment may not apply to the next person. This limits how far results can be generalised and makes it harder to build a robust evidence base for each application. This also bears on the program: because frequency bands are broad indicators rather than definitive markers of a mental state, a mapping tuned to one person's data cannot be assumed to work for another's.

In addition, ethical concerns have arisen because EEG data can reveal private cognitive information about the user, raising questions of mental privacy and consent when such data are stored or shared (Jwa et al., 2022). Users may also be reluctant to share such information with consumer devices (Farahany et al., 2018), especially when much of the early data would be collected for studies on effectiveness. Finally, this study has its own limitations: its conclusions rest mainly on existing literature, and its program was exploratory, did not use recorded EEG, and was not evaluated by listeners.

Future research should focus on making EEG sonification more accurate, affordable, and accessible. Improved wearable EEG devices and signal-processing techniques could allow for more reliable real-time feedback, while larger controlled studies could determine whether interventions such as neurofeedback produce lasting benefits. For brain-computer music

interfaces, reducing reliance on visual control could also make musical expression accessible to individuals with more severe disabilities. Machine-learning methods could also improve the accuracy of these systems, for example through automated artifact removal and subject-specific calibration of the mapping between EEG features and sound. For this project specifically, the next step would be to run the feature extraction on recorded EEG, connect band power to pitch or tempo, and test whether listeners can tell different brain states apart by ear.

Conclusion

EEG sonification offers a new way to transform invisible brain activity into something that can be heard, interpreted, and used. In response to the initial research question, the evidence suggests that EEG sonification can extend the use of EEG beyond traditional diagnosis into meaningful applications, although these applications remain at an early stage. Brain–computer music interfaces show clear potential for accessibility, giving people with severe motor disabilities a means of creative expression. Music-based rehabilitation also appears promising, though it requires further controlled research. Evidence for neurofeedback and wearable devices is the most mixed, with uncertain benefits and feasibility. Across all four applications, the main limitation is the variability and inconsistency of EEG signals between individuals. Frequency bands are difficult to generalize, which makes results difficult to interpret and draw conclusions from. EEG sonification is therefore best seen, at present, as a complement to existing diagnostic EEG technologies that extends their uses, rather than as a replacement for diagnosis and monitoring in clinical settings.

Data and Code Availability

The Python code used to generate the waveforms, envelopes, melody, and figures in this study is available at [repository URL or DOI, e.g., GitHub/Zenodo].

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